

Data Preprocessing

Data preprocessing is the process of preparing raw data for analysis by cleaning and transforming it into a usable format.

- In **data mining** it refers to preparing raw data for mining by performing tasks like cleaning, transforming, and organizing it into a format suitable for mining algorithms.
- Goal is to improve the quality of the data.
- Helps in handling missing values, removing duplicates, and normalizing data.
- Ensures the accuracy and consistency of the dataset.

Steps in Data Preprocessing

Some key steps in data preprocessing are **Data Cleaning**, **Data Integration**, **Data Transformation**, and **Data Reduction**.

1. Data Cleaning

- It is the process of identifying and correcting errors or inconsistencies in the dataset.
- It involves handling missing values, removing duplicates, and correcting incorrect or outlier data to ensure the dataset is accurate and reliable.
- Clean data is essential for effective analysis, as it improves the quality of results and enhances the performance of data models.

Missing Values:

- This occurs when data is absent from a dataset.
- You can either ignore the rows with missing data or fill the gaps manually, with the attribute mean, or by using the most probable value.
- This ensures the dataset remains accurate and complete for analysis.

Noisy Data:

- It refers to irrelevant or incorrect data that is difficult for machines to interpret, often caused by errors in data collection or entry. It can be handled in several ways:
 - **Binning Method:** The data is sorted into equal segments, and each segment is smoothed by replacing values with the mean or boundary values.

- Regression: Data can be smoothed by fitting it to a regression function, either linear or multiple, to predict values.
- Clustering: This method groups similar data points together, with outliers either being undetected or falling outside the clusters. These techniques help remove noise and improve data quality.

Removing Duplicates:

- It involves identifying and eliminating repeated data entries to ensure accuracy and consistency in the dataset.
- This process prevents errors and ensures reliable analysis by keeping only unique records.

2. Data Integration

- It involves merging data from various sources into a single, unified dataset. It can be challenging due to differences in data formats, structures, and meanings.
- Techniques like record linkage and data fusion help in combining data efficiently, ensuring consistency and accuracy.

Record Linkage:

- Record Linkage is the process of identifying and matching records from different datasets that refer to the same entity, even if they are represented differently.
- It helps in combining data from various sources by finding corresponding records based on common identifiers or attributes.

Data Fusion:

- Data Fusion involves combining data from multiple sources to create a more comprehensive and accurate dataset.
- It integrates information that may be inconsistent or incomplete from different sources, ensuring a unified and richer dataset for analysis.

3. Data Transformation

- It involves converting data into a format suitable for analysis.
- Common techniques include normalization, which scales data to a common range; standardization, which adjusts data to have zero mean and unit variance; and discretization, which converts continuous data into discrete categories.
- These techniques help prepare the data for more accurate analysis.
- Data Normalization: The process of scaling data to a common range to ensure consistency across variables.
- Discretization: Converting continuous data into discrete categories for easier analysis.
- Data Aggregation: Combining multiple data points into a summary form, such as averages or totals, to simplify analysis.
- Concept Hierarchy Generation: Organizing data into a hierarchy of concepts to provide a higher-level view for better understanding and analysis.

4. Data Reduction

- It reduces the dataset's size while maintaining key information.
- This can be done through feature selection, which chooses the most relevant features, and feature extraction which transforms the data into a lower-dimensional space while preserving important details.
- It uses various reduction techniques such as:
 - Dimensionality Reduction (e.g., Principal Component Analysis): A technique that reduces the number of variables in a dataset while retaining its essential information.
 - Numerosity Reduction: Reducing the number of data points by methods like sampling to simplify the dataset without losing critical patterns.
 - Data Compression: Reducing the size of data by encoding it in a more compact form, making it easier to store and process.

Uses of Data Preprocessing

Data preprocessing is utilized across various fields to ensure that raw data is transformed into a usable format for analysis and decision-making.

- Here are some key areas where data preprocessing is applied:

1. Data Warehousing:

- In data warehousing, preprocessing is essential for cleaning, integrating, and structuring data before it is stored in a centralized repository.
- This ensures the data is consistent and reliable for future queries and reporting.

2. Data Mining:

- Data preprocessing in data mining involves cleaning and transforming raw data to make it suitable for analysis.
- This step is crucial for identifying patterns and extracting insights from large datasets.

3. Machine Learning:

- In machine learning, preprocessing prepares raw data for model training.
- This includes handling missing values, normalizing features, encoding categorical variables, and splitting datasets into training and testing sets to improve model performance and accuracy.

4. Data Science:

- Data preprocessing is a fundamental step in data science projects, ensuring that the data used for analysis or building predictive models is clean, structured, and relevant.
- It enhances the overall quality of insights derived from the data.

5. Web Mining:

- In web mining, preprocessing helps analyze web usage logs to extract meaningful user behavior patterns.
- This can inform marketing strategies and improve user experience through personalized recommendations.

6. Business Intelligence (BI):

- Preprocessing supports BI by organizing and cleaning data to create dashboards and reports that provide actionable insights for decision-makers.

7. Deep Learning Purpose:

- Similar to machine learning, deep learning applications require preprocessing to normalize or enhance features of the input data, optimizing model training processes.

Advantages of Data Preprocessing

- Improved Data Quality: Ensures data is clean, consistent, and reliable for analysis.
- Better Model Performance: Reduces noise and irrelevant data, leading to more accurate predictions and insights.
- Efficient Data Analysis: Streamlines data for faster and easier processing.
- Enhanced Decision-Making: Provides clear and well-organized data for better business decisions.

Disadvantages of Data Preprocessing

- Time-Consuming: Requires significant time and effort to clean, transform, and organize data.
- Resource-Intensive: Demands computational power and skilled personnel for complex preprocessing tasks.
- Potential Data Loss: Incorrect handling may result in losing valuable information.
- Complexity: Handling large datasets or diverse formats can be challenging.

Data Cleaning

- Data cleaning is a step in machine learning (ML) which involves identifying and removing any missing, duplicate or irrelevant data.
- Raw data (log file, transactions, audio /video recordings, etc) is often noisy, incomplete and inconsistent which can negatively impact the accuracy of model.
- The goal of data cleaning is to ensure that the data is accurate, consistent and free of errors.
- Clean datasets also important in EDA (Exploratory Data Analysis) which enhances the interpretability of data so that the right actions can be taken based on insights.

*How to Perform **Data Cleaning** ?

The process begins by identifying issues like missing values, duplicates and outliers. Performing data cleaning involves a systematic process to identify and remove errors in a dataset.

The following steps are essential to perform data cleaning:

- Remove Unwanted Observations: Eliminate duplicates, irrelevant entries or redundant data that add noise.
- Fix Structural Errors: Standardize data formats and variable types for consistency.
- Manage Outliers: Detect and handle extreme values that can skew results, either by removal or transformation.
- Handle Missing Data: Address gaps using imputation, deletion or advanced techniques to maintain accuracy and integrity.

Implementation for Data Cleaning

Step1: Import Libraries and Load Dataset

Step 2: Check for Duplicate Rows

Step 3: Identify Column Data Types

Step 4: Count Unique Values in the Categorical Columns

Step 5: Calculate Missing Values as Percentage

Step 6: Drop Irrelevant or Data-Heavy Missing Columns

Step 7: Detect Outliers with Box Plot

Step 8: Calculate Outlier Boundaries and Remove Them

Step 9: Impute Missing Data Again if Any

Step 10: Recalculate Outlier Bounds and Remove Outliers from the Updated Data

Step 11: Data validation and verification

Step 12: Data formatting

Advantages:

- Improved model performance: Removal of errors, inconsistencies and irrelevant data helps the model to better learn from the data.
- Increased accuracy: Helps ensure that the data is accurate, consistent and free of errors.
- Better representation of the data: Data cleaning allows the data to be transformed into a format that better represents the underlying relationships and patterns in the data.
- Improved data quality: Improve the quality of the data, making it more reliable and accurate.
- Improved data security: Helps to identify and remove sensitive or confidential information that could compromise data security

Disadvantages:

- Time-consuming: It is very time-consuming task specially for large and complex datasets.
- Error-prone: It can result in loss of important information.
- Cost and resource-intensive: It is resource-intensive process that requires significant time, effort and expertise. It can also require the use of specialized software tools.
- Overfitting: Data cleaning can contribute to overfitting by removing too much data.

Handling Missing Values

- Missing values are a common challenge in machine learning and data analysis.
- They occur when certain data points are missing for specific variables in a dataset.
- These gaps in information can take the form of blank cells, null values or special symbols like "NA", "NaN" or "unknown."
- If not addressed properly, missing values can harm the accuracy and reliability of our models.
- They can reduce the sample size, introduce bias and make it difficult to apply certain analysis techniques that require complete data.
- Efficiently handling missing values is important to ensure our machine learning models produce accurate and unbiased results.

Importance of Handling Missing Values

- Handling missing values is important for ensuring the accuracy and reliability of data analysis and machine learning models. Key reasons include:
 - Improved Model Accuracy: Addressing missing values helps avoid incorrect predictions and boosts model performance.
 - Increased Statistical Power: Imputation or removal of missing data allows the use of more analysis techniques, maintaining the sample size.
 - Bias Prevention: Proper handling ensures that missing data doesn't introduce systematic bias, leading to more reliable results.
 - Better Decision-Making: A clean dataset leads to more informed, trustworthy decisions based on accurate insights.

Challenges Posed by Missing Values

- Missing values can introduce several challenges in data analysis including:
 - Reduce sample size: If rows or data points with missing values are removed, it reduces the overall sample size which may decrease the reliability and accuracy of the analysis.
 - Bias in Results: When missing data is not handled carefully, it can introduce bias. This is especially problematic when the missingness is not random, leading to misleading conclusions.
 - Difficulty in Analysis: Many statistical techniques and machine learning algorithms require complete data for all variables. Missing values can cause certain analyses or models to be inapplicable, limiting the methods we can use.

Reasons Behind Missing Values in the Dataset

- Data can be missing from a dataset for several reasons and understanding the cause is important for selecting the most effective way to handle it. Common

Reasons for missing data include:

- Technical issues: Failed data collection or errors during data transmission.
- Human errors: Mistakes like incorrect data entry or oversights during data processing.
- Privacy concerns: Missing sensitive or personal information due to confidentiality policies.
- Data processing issues: Errors that occur during data preparation.

- By identifying the reason behind the missing data, we can better assess its impact whether it's causing bias or affecting the analysis and select the proper handling method such as imputation or removal.

Types of Missing Values

- Missing values in a dataset can be categorized into three main types each with different implications for how they should be handled:

- **Missing Completely at Random (MCAR):** In this case, the missing data is completely random and unrelated to any other variable in the dataset.

- The absence of data points occurs without any systematic pattern such as a random technical failure or data omission.

- **Missing at Random (MAR):** The missingness is related to other observed variables but not to the value of the missing data itself.

For example, if younger individuals are more likely to skip a particular survey question, the missingness can be explained by age but not by the content of the missing data.

- **Missing Not at Random (MNAR):** Here, the probability of missing data is related to the value of the missing data itself.

For example, people with higher incomes may be less likely to report their income, leading to a direct connection between the missingness and the value of the missing data.

Representation of Missing Values in Datasets

- Missing values can be represented by blank cells, specific values like "NA" or codes.
- It's important to use consistent and documented representation to ensure transparency and ease in data handling.
- Common representations include:
 - **Blank Cells:** Empty cells in data tables or spreadsheets are used to signify missing values. This is common in many data formats like CSVs.
 - **Specific Values:** It is commonly used placeholders for missing data include "NA", "NaN", "NULL" or even arbitrary values like 999. It's important to choose a standardized value and document its meaning to prevent confusion.

- **Codes or Flags:** In some cases, non-numeric codes or flags (e.g "MISSING", "UNKNOWN") are used to show missing data.
- These can be useful in distinguishing between different types of missingness or categorizing missing data based on its origin

Strategies for Handling Missing Values in Data Analysis

1. Removing Rows with Missing Values

- Removing rows with missing values is a simple and straightforward method to handle missing data, used when we want to keep our analysis clean and minimize complexity.

Advantages:

- Simple and efficient: It's easy to implement and quickly removes data points with missing values.
- Cleans data: It removes potentially problematic data points, ensuring that only complete rows remain in the dataset.

Disadvantages:

- Reduces sample size: When rows are removed, the overall dataset shrinks which can affect the power and accuracy of our analysis.
- Potential bias: If missing data is not random (e.g if certain groups are more likely to have missing values) removing rows could introduce bias.

2. Imputation Methods

- Imputation involves replacing missing values with estimated values.
- This approach is beneficial when we want to preserve the dataset's sample size and avoid losing data points.
- However, it's important to note that the accuracy of the imputed values may not always be reliable.

2.1 Mean, Median and Mode Imputation:

- This method involves replacing missing values with the mean, median or mode of the relevant variable.
- It's a simple approach but it doesn't account for the relationships between variables.

Advantages:

- Simple and efficient: Easy to implement and quick.
- Works well with numerical data: It is useful for numerical variables with a normal distribution.

Disadvantages:

- Inaccuracy: It assumes the missing value is similar to the central tendency (mean/median/mode) which may not always be the case.

2.2 Forward and Backward Fill

- Forward and backward fill techniques are used to replace missing values by filling them with the nearest non-missing values from the same column.
- This is useful when there's an inherent order or sequence in the data

Advantages:

- Simple and Intuitive: Preserves the temporal or sequential order in data.
- Preserves Patterns: Fills missing values logically, especially in time series or ordered data.

Disadvantages:

- Assumption of Closeness: Assumes that the missing values are similar to the observed values nearby which may not always be true.
- Potential Inaccuracy: May not work well if there are large gaps between non-missing values.

Note:

- Forward fill uses the last valid observation to fill missing values.
- Backward fill uses the next valid observation to fill missing values.

3. Interpolation Techniques

- Interpolation is a technique used to estimate missing values based on the values of surrounding datapoints.
- Unlike simpler imputation methods (e.g mean, median, mode), interpolation uses
- the relationship between neighboring values to make more informed estimations.
- The interpolate() method in pandas are divided into Linear and Quadratic.

Advantages:

- **Sophisticated Approach:** Interpolation is more accurate than simple imputation methods like mean or median, as it considers the underlying data structure.
- **Preserves Data Relationships:** Captures patterns or trends that exist between data points, which helps maintain the integrity of the dataset.

Disadvantages:

- **Complexity:** Requires more computational resources and additional libraries.
- **Assumptions on Data:** Assumes that data points follow a specific pattern (e.g., linear or quadratic), which may not always be true.

Note:

- Linear interpolation assumes a straight line between two adjacent non-missing values.
- Quadratic interpolation assumes a quadratic curve that passes through three adjacent non-missing values.

Detect and Remove the Outliers

- Outliers are data points that deviate significantly from other data points in a dataset.
- They can arise from a variety of factors such as measurement errors, rare events or natural variations in the data.
- If left unchecked it can distort data analysis, skew statistical results and impact machine learning model performance. **Common Causes of Outliers**

- Understanding the causes of outliers helps in finding the best approach to handle them. Some common causes include:

- Measurement errors: Errors during data collection or from instruments can result in extreme values that don't reflect the underlying data distribution.
- Sampling errors: Outliers can arise if the sample we collected isn't representative of the population we're studying.
- Natural variability: Certain data points naturally fall outside the expected range especially in datasets with inherently high variability.
- Data entry errors: Mistakes made during manual data entry such as incorrect values or typos can create outliers.
- Experimental errors: Outliers can occur due to equipment malfunctions, environmental factors or unaccounted variables in experiments.
- Sampling from multiple populations: Combining data from distinct populations with different characteristics can create outliers if researchers don't properly segment the datasets.
- Intentional outliers: Sometimes outliers are deliberately introduced into datasets for testing purposes to evaluate the robustness of models or algorithms.

Need for Outliers Removal

Outliers can create significant issues in data analysis and machine learning which makes their removal important:

- Skewed Statistical Measures: Outliers can distort the mean, standard deviation and correlation values.

For example, an extreme value can make the mean unrepresentative of the actual data which leads to incorrect conclusions.

Reduced Model Accuracy:

- Outliers can influence machine learning models especially those sensitive to extreme values like linear regression. They may cause the model to focus too much on these rare events helps in reducing its ability to generalize to new, unseen data.

Misleading Visualizations:

- Outliers can stretch the scale of charts and graphs help in making it difficult to interpret the main data trends.

For example, when visualizing a dataset with a few extreme values, it might find meaningful patterns in the majority of the data.

- By removing or handling outliers, we prevent these issues and ensure more accurate analysis and predictions. Methods for Detecting and Removing Outliers
- There are several ways to detect and handle outliers in Python.
- We can use visualization techniques or statistical methods depending on the nature of our data Each method serves different purposes and is suited for specific types of data.

Methods for Detecting and Removing Outliers

- There are several ways to detect and handle outliers in Python.
 - We can use visualization techniques or statistical methods depending on the nature of our data Each method serves different purposes and is suited for specific types of data.
 - Here we will be using Pandas and Matplotlib Diabetes dataset which is preloaded in the Sckit libraries on the-learn library.
- 1. Visualizing and Removing Outliers Using Box Plots**
 - 2. Visualizing and Removing Outliers Using Scatter Plots**
 - 3. Z-Score Method for Outlier Detection**
 - 4. Interquartile Range (IQR) Method**

What is Data Transformation?

- Data transformation is an important step in data analysis process that involves the conversion, cleaning, and organizing of data into accessible formats.
- It ensures that the information is accessible, consistent, secure, and finally recognized by the intended business users.
- This process is undertaken by organizations to utilize their data to generate timely business insights and support decision-making processes.

The transformations can be divided into two categories:

- Simple Data Transformations involve basic tasks like cleansing, standardization, aggregation, and filtering used to prepare data for analysis or reporting through straightforward manipulation techniques
- Complex Data Transformations involve advanced tasks like integration, migration, replication, and enrichment. They require techniques such as data modeling, mapping, and validation, and are used to prepare data for machine learning, advanced analytics, or data warehousing, Importance of Data Transformation
- Data transformation is important because it improves data quality, compatibility, and utility.
- The procedure is critical for companies and organizations that depend on data to make informed decisions because it assures the data's accuracy,
- reliability, and accessibility across many systems and applications.
- Improved Data Quality: Data transformation eliminates mistakes, inserts in missing information, and standardizes formats, resulting in higher-quality, more dependable, and accurate data.
- Enhanced Compatibility: By converting data into a suitable format, companies may avoid possible compatibility difficulties when integrating data from many sources or systems.
- Simplified Data Management: Data transformation is the process of evaluating and modifying data to maximize storage and discoverability, making it simpler to manage and maintain.
- Broader Application: Transformed data is more useable and applicable in a larger variety of scenarios, allowing enterprises to get the most out of their data.
- Faster Queries: By standardizing data and appropriately storing it in a warehouse, query performance and BI tools may be enhanced, resulting in less friction during analysis

Data Transformation Techniques and Tools

- There are several ways to alter data, including:
- **Programmatic Transformation:** Automated using scripts in Python, R, or SQL.
- **ETL Tools:** Automate extract-transform-load processes for large scale data (e.g., Talend, Informatica).

- **Normalization/Standardization:** Use Min-Max Scaler, Standard Scaler from Scikit-learn.
- **Categorical Encoding** such as **One-hot get dummies** (Pandas) and Label Encoder(Scikit-learn)
- fillna() (Pandas)
- Simple Imputer (Scikit-learn) for mean/median/mode imputation.
- **Feature Engineering:** Create using apply, map, transform in Pandas. new features
- **Aggregation/Grouping:** Use group by in Pandas for sum, mean, count, etc.
- **Text Preprocessing:** Tokenization, stemming, stop-word removal using NLTK and SpaCy.
- **Dimensionality Reduction:** learn's PCA and TruncatedSVD.

Applications of Data Transformation

- Applications for data transformation are found in a number of industries:
- Business intelligence (BI) is the process of transforming data for use in real-time reporting and decision-making using BI technologies.
- Healthcare: Ensuring interoperability across various healthcare systems by standardization of medical records.
- Financial Services: Compiling and de-identifying financial information for reporting and compliance needs.
- Retail: Improving customer experience through data transformation into an analytics-ready format and customer behavior analysis.
- Customer Relationship Management (CRM): By converting customer data, firms may obtain insights into consumer behavior, tailor marketing strategies, and increase customer satisfaction

Feature Engineering:

Scaling, Normalization and Standardization

- Feature engineering is the process of creating, transforming or selecting the most relevant variables (features) from raw data to improve model performance.

- Effective features help the model capture important patterns and relationships in the data.
- It directly contributes to model building in the following ways:
- Well-designed features allow models to learn complex patterns more effectively.
- Reduces noise and irrelevant information, improving prediction accuracy.
- Helps prevent overfitting by emphasizing meaningful data signals.
- Simplifies model interpretation by creating more informative and understandable inputs.
- There are various techniques, such as scaling, normalization and standardization that can be used for feature engineering. Let's explore some of them.

1. Absolute Maximum Scaling

- Absolute Maximum Scaling rescales each feature by dividing all values by the maximum absolute value of that feature.
- This ensures the feature values fall within the range of -1 to 1.
- While simple and useful in some contexts, it is highly sensitive to outliers which can skew the max absolute value and negatively impact scaling quality.

2. Min-Max Scaling

- Min-Max Scaling transforms features by subtracting the minimum value and dividing by the difference between the maximum and minimum values.
- This method maps feature values to a specified range, commonly 0 to 1, preserving the original distribution shape but is still affected by outliers due to reliance on extreme values

3. Normalization (Vector Normalization)

- Normalization scales each data sample (row) such that its vector length (Euclidean norm) is 1.

- This focuses on the direction of data points rather than magnitude making it useful in algorithms where angle or cosine similarity is relevant, such as text classification or clustering.

4. Standardization

- Standardization centers features by subtracting the mean and scales them by dividing by the standard deviation, transforming features to have zero mean and unit variance.
- This assumption of normal distribution often benefits models like linear regression, logistic regression and neural networks by improving convergence speed and stability

5. Robust Scaling

- Robust Scaling uses the median and interquartile range (IQR) instead of the mean and standard deviation making the transformation robust to outliers and skewed distributions.
- It is highly suitable when the dataset contains extreme values or noise

Let's see the key differences across the five main feature scaling techniques commonly used in machine learning preprocessing.

Type	Method Description	Sensitivity to Outliers	Typical Use Cases
Absolute Maximum Scaling	Divides values by max absolute value in each feature	High	Sparse data, simple scaling
Min-Max Scaling (Normalization)	Scales features to by min-max normalization	High	Neural networks, bounded input features
Normalization (Vector Norm)	Scales each sample vector to unit length (norm = 1)	Not applicable (per row)	Direction-based similarity, text classification
Standardization (Z-Score)	Centers features to mean 0 and scales to unit variance	Moderate	Most ML algorithms, assumes approx. normal data
Robust Scaling	Centers on median and scales using IQR	Low	Data with outliers, skewed distributions

Advantages

- **Improves Model Performance:** Enhances accuracy and predictive power by presenting features in comparable scales.
- **Speeds Up Convergence:** Helps gradient-based algorithms train faster and more reliably.
- **Prevents Feature Bias:** Avoids dominance of large-scale features, ensuring fair contribution from all features.
- **Increases Numerical Stability:** Reduces risks of overflow/underflow in computations.
- **Facilitates Algorithm Compatibility:** Makes data suitable for distance- and gradient-based models like SVM, KNN and neural networks

What is Data Sampling - Types, Importance, Best Practices

- Data sampling is a statistical method that selects a representative subset (sample) from a large dataset.
- Analysts then study this sample to make inferences and draw conclusions about the entire dataset.
- It's a powerful tool for handling large volumes of data efficiently

Data Sampling Process

- The process of data sampling involves the following steps:
- **Identify the Target Dataset:** Choose the large dataset you want to study. it represents the whole population.
- **Determine Sample Size:** Decide how many data points to include in your sample. This subset will be used for analysis
- **Choose a Sampling Method:** Select a suitable technique like Random, Systematic, Cluster, Stratified, or Snowball sampling, based on your goals and data type.
- **Collect the Sample:** Apply the chosen method to extract the sample from the dataset systematically.
- **Analyze the Sample:** Study the sample to understand patterns and characteristics using statistical tools.
- **Generalize to the Population:** Use insights from the sample to make predictions or conclusions about the entire dataset

Types of Data Sampling Techniques

There are mainly two types of Data Sampling techniques which are further divided into 4 sub-categories each.

They are as follows:

1. Probability Data Sampling Technique

- Probability Sampling ensures every data point has a known, nonzero chance of being selected. This helps create a representative sample, allowing reliable generalization to the whole population.
- Simple Random Sampling: Each data point has an equal chance of selection. Example: Tossing a coin—head or tail has equal probability.
- Systematic Sampling: Data is selected at regular intervals from an ordered list. Example: From 10 entries, selecting every 2nd one (2nd, 4th, 6th...).
- Stratified Sampling: Data is divided into groups (strata) based on shared traits, and samples are drawn from each group. Example: Dividing employees by gender, then sampling from each group.
- Cluster Sampling: Random groups (clusters) are selected, and then all or some members within them are sampled. Example: Choosing random user groups from different mobile networks.

2. Non-Probability Data Sampling

- Non-probability data sampling means that the selection happens on a non-random basis, and it depends on the individual as to which data does it want to pick. There is no random selection, and every selection is made by a thought and an idea behind it.
- Convenience Sampling: Data is selected based on ease of access and minimal effort. Example: Choosing the most recent or easily available IT recruitment data.
- Voluntary Response Sampling: Participants choose to take part in the study on their own. Example: A blood group survey where only willing participants respond.
- Purposive Sampling: Data is selected for a specific purpose or characteristic. Example: Surveying rural areas to study educational needs.
- Snowball Sampling: Participants recruit others, growing the sample like a snowball. Example: One slum resident leads to another in a housing conditions survey

Advantage and Disadvantage

Advantage	Disadvantage
Helps draw conclusion about large datasets from smaller samples.	Difference between sample and population reduces accuracy
saves times with faster data analysis	Difficulties in some methods like cluster formation
cost effective reduces expenses in data collection and processing	Sample may not represent the population due to poor sampling technique
can produce accurate and reliable result when sampling is done correctly	Inaccurate conclusions if the sample lacks proper representation or is too small